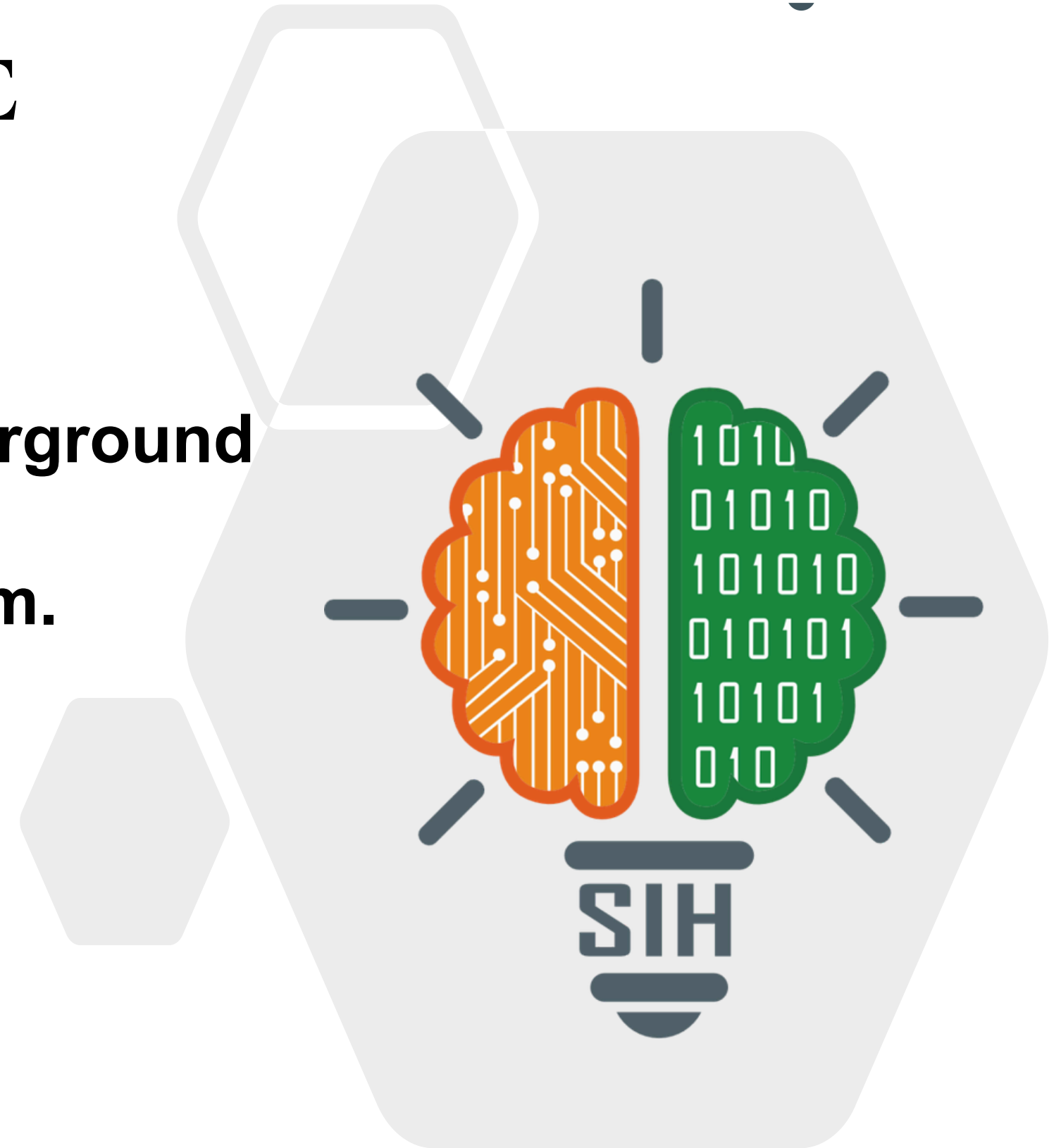
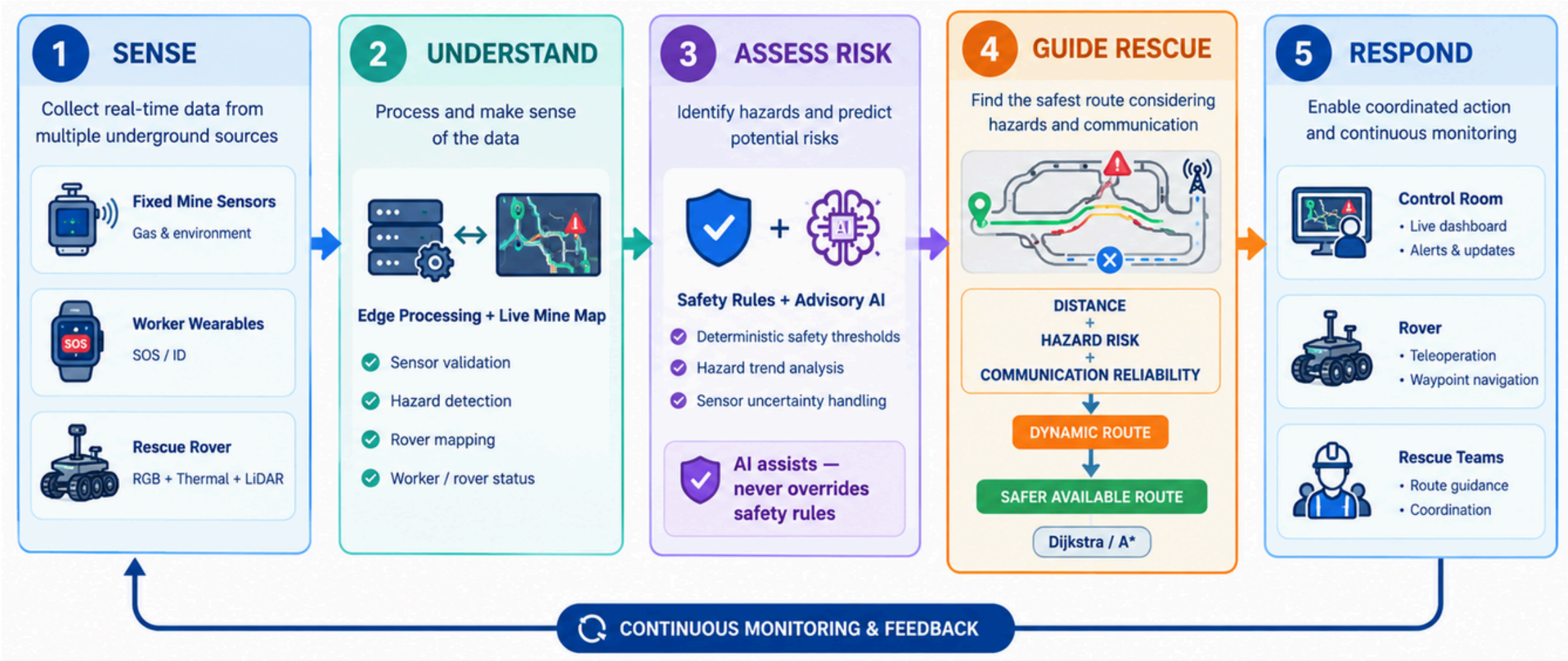


TITLE PAGE

- Problem Statement ID –SIH26039
- Problem Statement Title- AI-Powered Underground Mine Safety, Monitoring and Rescue System.
- Theme-Smart Automation
- PS Category-Hardware
- Team ID-
- Team Name - ShaktiMine





WHAT MAKES OUR APPROACH DIFFERENT?

COMMUNICATION-AWARE RESCUE ROUTING

We don't choose a route based on distance alone. The system considers hazard risk + distance + communication reliability to guide rescue through the safest available connected path.

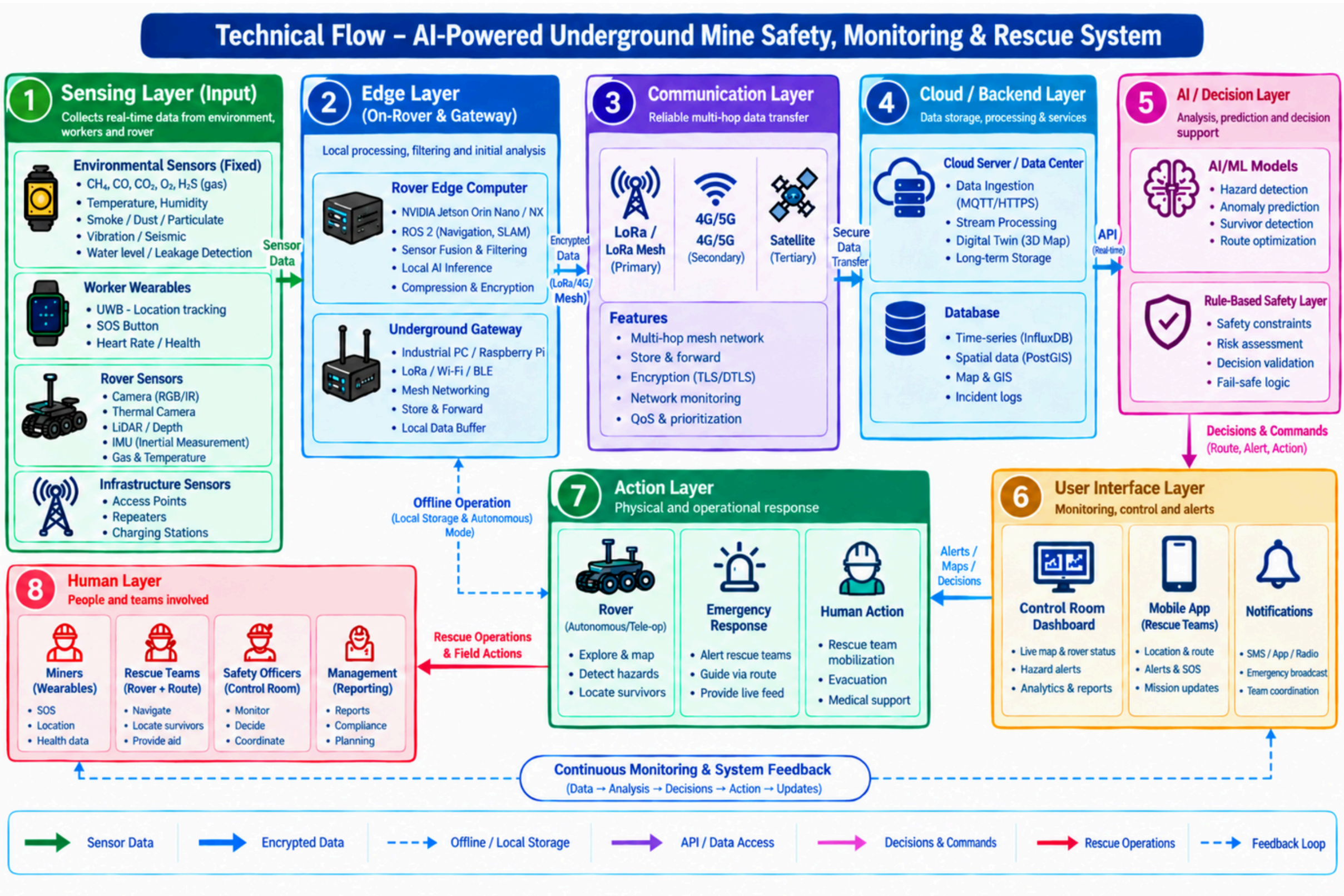
AI THAT CANNOT OVERRIDE SAFETY

AI assists decisions it does not control safety. AI detects trends and provides recommendations, while deterministic safety rules remain the final constraint.

GRACEFUL DEGRADATION

Safety doesn't depend on every component working perfectly. Sensor, communication, rover, backend or AI failures trigger fallback mechanisms instead of assuming conditions are safe.

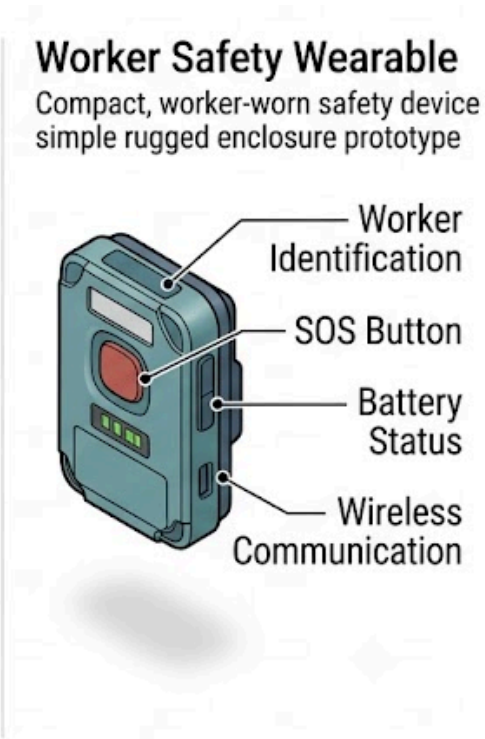
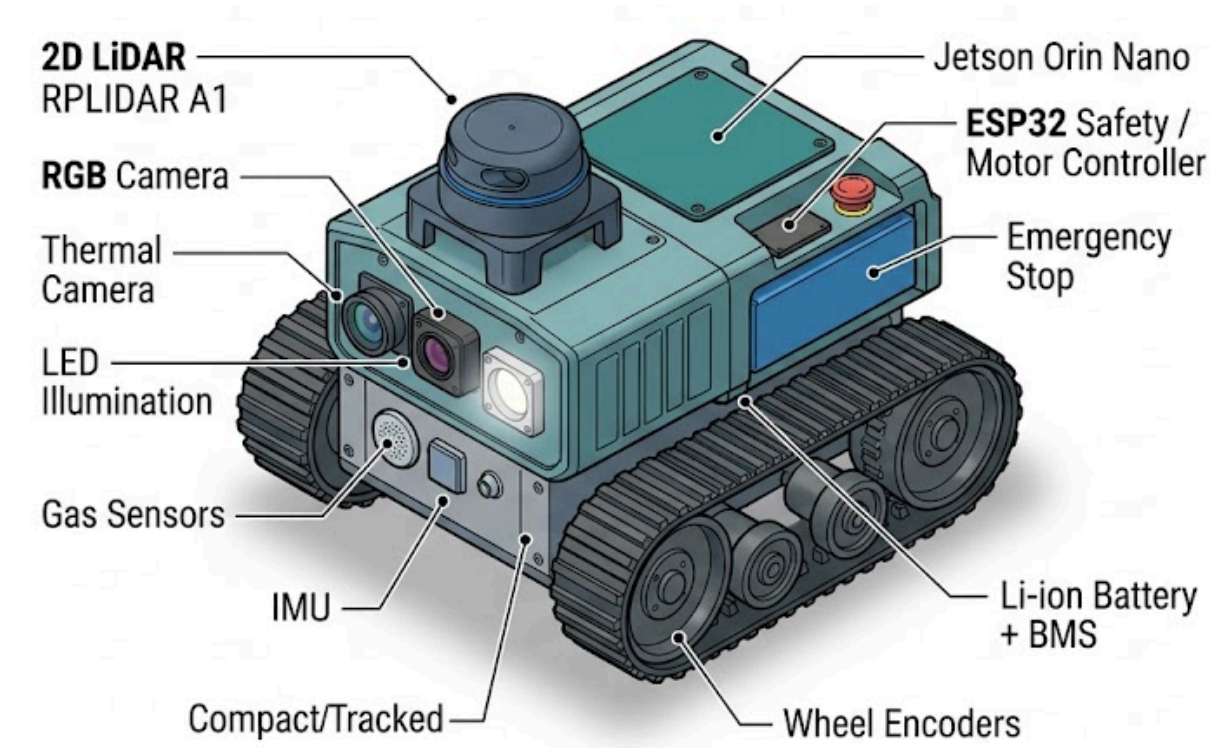
Problem	How Our Solution Addresses It
Hazardous gases & air conditions	Continuous gas/environment monitoring with deterministic hazard alerts
Unknown underground conditions	Rover reconnaissance using RGB, thermal & LiDAR
Worker emergencies	Wearable SOS + worker identification
Unsafe rescue routes	Dynamic routing using distance + hazard risk + communication reliability
Communication loss	LoRa mesh + store-and-forward for resilient telemetry
System / sensor failures	Validation + uncertainty states + fallback sensing instead of assuming safety



Category	Technologies / Components
Programming	Python • C/C++ • JavaScript
Embedded Hardware	ESP32 / STM32 • Jetson Orin Nano / Raspberry Pi
Sensors	CH ₄ • CO • O ₂ • Temperature/Humidity •
Rover Sensors	LiDAR • RGB Camera • Thermal Camera • IMU • Wheel
Communication	LoRa Mesh • Wi-Fi • BLE • MQTT
Robotics	ROS 2 • SLAM Toolbox • Nav2
AI / ML	OpenCV • YOLO-based Detection • EWMA / Anomaly
Backend & Database	FastAPI • PostgreSQL / TimescaleDB
Routing	Dijkstra / A*
Dashboard	React / Next.js • REST API • WebSocket
Security	TLS • HMAC • JWT • Role-Based Access

Practical prototype | Controlled scope | Risk-aware design

TECHNOLOGICAL	ECONOMIC	SOCIAL
Proven hardware + software stack	₹35K–₹72K prototype	Rover-first reconnaissance
1 Rover + 3 Nodes + 2 Wearables	Off-the-shelf components	Worker SOS + identification
LoRa + AI + deterministic safety	Controlled Prototype Scope 1 rover + 3 fixed nodes + 2 wearables	Human-in-the-loop + fallback safety



Risk	Mitigation
Communication loss	LoRa mesh + store-and-forward + last-known timestamp
LiDAR affected by dust	Protective shroud/wiper + sensor fault detection
Sensor drift/stuck value	Validation + filtering + “UNKNOWN” state
Rover unavailable	Fixed sensors + worker wearables provide fallback coverage
AI failure	Deterministic safety rules operate independently
Conflicting readings	Flag uncertainty; never average into a false-safe decision
Deployment safety	Prototype remains surface/mockup only ; future deployment requires certified Exd/Exi hardware and DGMS compliance

SOCIAL

Protect people first



Underground mine worker Worker safety wearable SOS Rescue rover Rescue team

Rover gathers information before rescuers are exposed to uncertain conditions

- Reduced human exposure during reconnaissance
- Worker SOS & identification
- Human-in-the-loop rescue decisions
- Better situational awareness

OPERATIONAL

Reduce uncertainty



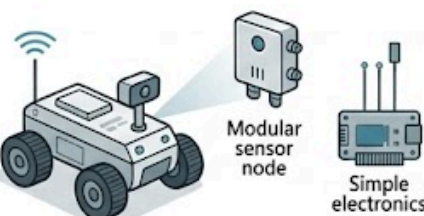
Fixed monitoring points Live route/map SAFER AVAILABLE ROUTE

DISTANCE + HAZARD RISK + COMMUNICATION RELIABILITY

- Continuous hazard monitoring
- Communication-resilient telemetry
- Risk-aware route planning
- Graceful degradation during failures

ECONOMIC

Buildable & scalable



Modular sensor node Simple electronics

Clean engineering prototype concept, (not 'guaranteed safest')

- Low-cost prototype architecture
- Open, vendor-neutral integration
- Modular hardware
- Incremental deployment

FROM UNCERTAINTY → INFORMED RESPONSE

BEFORE	WITH OUR SYSTEM
Human exposure during initial reconnaissance	Rover-first reconnaissance before rescue-team entry
Limited visibility into underground conditions	RGB + Thermal + LiDAR situational sensing
Fragmented information from different sources	3 sensing sources integrated into one live mine view
Routes planned under uncertainty	Dynamic routing: Distance + Hazard Risk + Communication Reliability
Communication loss → information gap	Mesh + Store-and-Forward preserves last-known state
Failure can create false confidence	Unknown / Uncertain state + rule-based safety layer

RESEARCH & FIELD EVIDENCE

Documented mining safety risks • Industry scale • Evidence behind the problem

DGMS • ANNUAL REPORT 2024

33%

Approximately 33% of fatal accidents in belowground operations were attributed to roof/side-fall accidents in 2024.

Source: Directorate General of Mines Safety (DGMS), Annual Report 2024

DGMS • 2024

8

Fatalities from ground-movement accidents in Indian coal mines in 2024.

Source: Directorate General of Mines Safety (DGMS), Annual Report 2024

MINISTRY OF COAL • FY 2024-25

31.858 MT

Underground coal production in India

Source: Ministry of Coal, Coal Directory of India, FY 2024-25

MINISTRY OF COAL • FY 2024-25

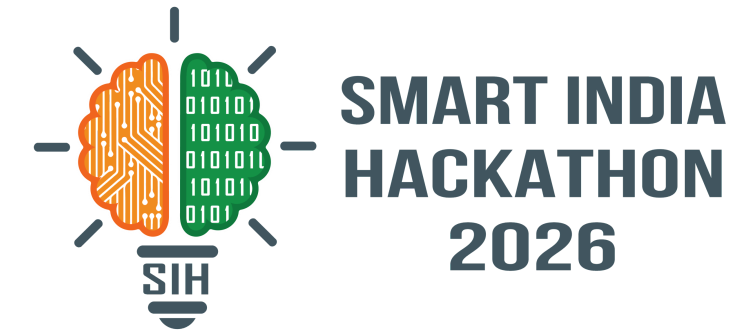
1,047.523 MT

Total coal production in India

Source: Ministry of Coal, Coal Directory of India, FY 2024-25



RESEARCH AND REFERENCES



[1] Directorate General of Mines Safety, Annual Report 2024, Ministry of Labour & Employment, Government of India, Dhanbad, 2024. [Online].

Available:

https://dgms.gov.in/writereaddata/UploadFile/AR_2024_Final_16122025.pdf

[2] Directorate General of Mines Safety, SANKET: Statistical Analysis & Numbers — Key Evaluation of Trends in Coal Mine Accidents, 2016–2022, Ministry of Labour & Employment, Government of India, Dhanbad, Feb. 2024. [Online].

Available:

https://dgms.gov.in/writereaddata/UploadFile/sanket0404_2024.pdf

[3] Ministry of Coal, Government of India, Coal Directory of India 2024–25, Office of the Deputy Director General, New Delhi, 2025. [Online].

Available:

<https://coal.gov.in/sites/default/files/2024-03/28-11-2025a-stc.pdf>

[4] L. K. Bandopadhyay, S. K. Chaulya, and P. K. Mishra, "Wireless information and safety system for mines," Journal of Scientific & Industrial Research, vol. 68, no. 2, pp. 107–117, 2009.

[5] CSIR–Central Institute of Mining and Fuel Research, "Wireless information and safety system for mines," U.S. Patent 8 587 414 B2, Nov. 19, 2013.

[6] P. R. Kumar (EasyM2M Technologies), quoted in "WiFi and wearable tracking devices for underground coal miners in India," International Mining, Jun. 19, 2015. [Online].

Available:

<https://im-mining.com/?p=28914>

[7] "India-based EasyM2M introduces underground WiFi facility for Eastern Coalfields mines," Mining Technology, 2015. [Online].

Available:

<https://www.mining-technology.com/news/newsindias-easym2m-introduces-underground-wifi-facility-for-eastern-coalfields-mines-4602814>

[8] Sandia National Laboratories, "Sandia Labs' Gemini-Scout robot likely to reach trapped miners ahead of rescuers," news release, Albuquerque, NM, Aug. 16, 2011. [Online].

Available:

<https://newsreleases.sandia.gov/?p=3499>